

Capstone Build

Project: Verified Study Buddy

1. Your mission

Your learning target

I can explain the mechanism and terminology of capstone grounded assistant.

I can apply the stated procedure to a new case with a traceable result.

I can identify an assumption or limitation and design a revealing follow-up.

Remember before reading

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

Without opening the old lesson, explain one key idea from Capstone Design. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Safety Boundaries. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Privacy by Design. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the evidence shows was wrong.

Week 34: Capstone Build

Implementation should preserve traceability from requirement to feature.



CREATOR MISSION

Build the smallest complete Study Buddy workflow.

Creator name and date:

Write or draw your reasoning and evidence.

Readiness for the connected explanation

Recall Capstone requirements and corpus. Explain why a recorded result needs its input conditions and evidence source before it can support a claim.

This week's end goal

By the end, I can explain capstone build using a traced example and a stated limitation.

2. Notice the evidence

PICTURE STORY

Notice → Predict → Explain

Begin with visible evidence. Do not explain more than the picture and information support.



MYSTERY

The learners must use capstone build to solve a problem. What information do they have? What decision or calculation do they make? What would count as evidence that it worked?

Three observations (what you can point to):

Write or draw your reasoning and evidence.

My prediction and the clue supporting it:

Write or draw your reasoning and evidence.

A closer explanation

Build the assistant as a sequence of testable stages: validate request, retrieve authorised evidence, decide answerability, draft, validate and return. Avoid merging all errors into one model score.

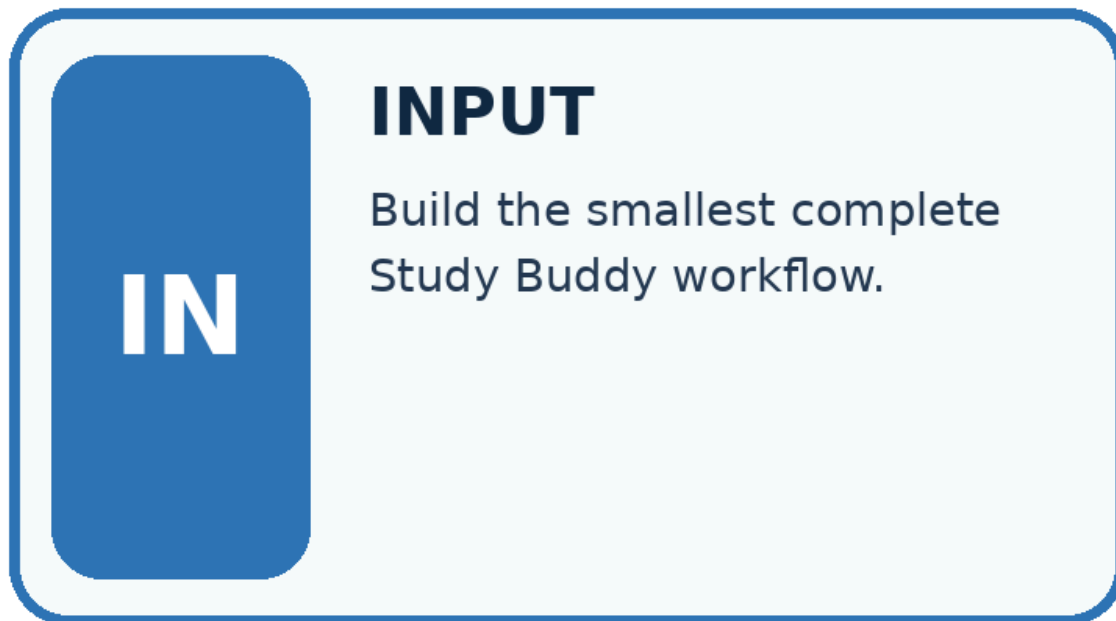
Begin by writing the question the procedure is intended to answer. Identify the information already supplied, the information it will produce and the information still missing. These are constructed classroom examples; they do not require live accounts, private records or an external service. Do not treat a plausible explanation as a measured outcome.

3. See how it works

VISUAL MECHANISM

Input → Process → Output

Trace the information. At each stage, ask what changed and how you could test it.



↓



SYSTEM

Implementation should preserve traceability from requirement to feature.





EVIDENCE

Build the smallest complete Study Buddy workflow.

Which stage could fail even when the other two are correct? Explain:

Write or draw your reasoning and evidence.

A closer explanation

A supported response requires current matching approval and register records. Any retrieval or validation failure gives needs_review with null claims. The classroom implementation performs no booking or external action.

Capstone grounded assistant — evidence trace

Input and scope · Check or calculation · Conclusion and limit

Input and scope: Request room B; retrieved approval names B but register names C. Draft gives capacity 10 from C.

Check or calculation: A supported response requires current matching approval and register records. Any retrieval or validation failure gives needs_review with null claims. The classroom implementation performs no booking or external action.

Conclusion and limit: Same-room support fails before or during semantic validation. Return needs_review with null claims; the draft capacity is unsupported for B.

The figure connects this chapter's worked input, declared procedure and result. Read the complete worked reasoning for qualifiers and intermediate steps; the abbreviated labels do not replace that reasoning.

Before applying the procedure, write its units, denominator or decision boundary where relevant. A partner should be able to repeat each step using the same inputs. If a required input is absent or violates the stated contract, record that condition explicitly instead of quietly replacing it with a convenient value.

Connect the picture and the explanation

Point to the input, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

4. Compare the cases

CASE GALLERY

Four Tests, Four Kinds of Evidence

A system is not understood until it survives more than one comfortable example.



NORMAL
Expected learner request



AMBIGUOUS
Two plausible meanings

NORMAL CASE: predict and name evidence. BOUNDARY CASE: predict and name evidence.



MISSING SOURCE

Evidence unavailable



ADVERSARIAL

Rules or trust challenged

MISSING INFORMATION: predict and name evidence. TRANSFER CASE: predict and name evidence.

Which case is most likely to reveal a hidden assumption? Why?

Write or draw your reasoning and evidence.

Change one thing in the pictured case

Change: E3 supplies losses of 1, 2 and 3 mL.

Prediction: Report those individual losses, not “half” unless starting volumes are also available.

Reason: Absolute losses alone do not establish each cup’s fractional loss.

5. Words that unlock the idea

VOCABULARY LAB

Words You Can Use to Reason

A definition matters when you can recognize the idea, use it, and contrast it with a near-miss.

1

TRACEABILITY

link requirement to design,
feature, and test

2

INTEGRATION

connect components into
one flow

3

VERSION CONTROL

record meaningful changes

4

MINIMUM COMPLETE

smallest end-to-end usable
system

Word · My own example · Not this / boundary

Word: traceability

My own example: The ability to connect a result back to its inputs, sources, steps and version.

Not this / boundary: _____

Word: integration

My own example: Connecting components so that they work together through agreed interfaces.

Not this / boundary: _____

Word: version control

My own example: A method for recording and comparing changes to identified versions of files.

Not this / boundary: _____

Word: minimum complete product

My own example: The smallest working version that fulfils its essential user task, including failure handling.

Not this / boundary: _____

Words used in the closer explanation

Word · Meaning

Word: Pipeline stage

Meaning: One operation with defined inputs and failure handling

Word: Answerability gate

Meaning: A check for sufficient evidence before answering

Word: Semantic validator

Meaning: A check that claims match task evidence

6. Follow a worked example

WORKED EVIDENCE

Reason Step by Step

Predict first. Then check each step. A correct answer without visible reasoning is incomplete evidence.

WEEK 34 • WORKED EVIDENCE

- 1 Requirement R1: answer only from approved sources.
- 2 Feature: retriever plus source-bound prompt and citation field.
- 3 Test E4 checks missing evidence triggers abstention.
- 4 Build one complete path before adding optional interfaces or media.

1. Requirement R1: answer only from approved sources.
2. Feature: retriever plus source-bound prompt and citation field.
3. Test E4 checks missing evidence triggers abstention.

4. Build one complete path before adding optional interfaces or media.

Recalculate, retrace, or restate the reasoning in your own representation:

Write or draw your reasoning and evidence.

CHECK Did the evidence answer the exact question? Did the total, path, or source record stay consistent?

A closer explanation

Request room B; retrieved approval names B but register names C. Draft gives capacity 10 from C.

Same-room support fails before or during semantic validation. Return needs_review with null claims; the draft capacity is unsupported for B.

Compare the intermediate evidence with the final statement. Explain why each number or decision follows from the stated inputs. The conclusion belongs to this scenario and procedure; changing the inputs, evaluation population or authority can change what it establishes. Retain the raw record so a later revision can be compared honestly.

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Capstone Build

Build a claim checker for a draft summary of a fictional class investigation.

The facts we will use

Approved record E2: three cups were observed for four days. No measured water loss is provided. Draft says “three cups were observed for four days and all lost half their water”.

All inputs and outcomes in this worked example are invented classroom data; no real model run is claimed.

Capstone Build

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Extract claims

Separate cup count, observation duration and the claimed water loss.



2. Match evidence

E2 supports three cups and four days; it supplies no water-loss result.



3. Repair draft

Retain the supported procedure facts; state that water-loss results were not supplied.

CHANGE ONE THING

E3 supplies losses of 1, 2 and 3 mL.

Explain the changed result using the matching lesson.

Read the picture one step at a time

Step · What happens and why

Step: 1. Extract claims

What happens and why: Separate cup count, observation duration and the claimed water loss.

Step: 2. Match evidence

What happens and why: E2 supports three cups and four days; it supplies no water-loss result.

Step: 3. Repair draft

What happens and why: Retain the supported procedure facts; state that water-loss results were not supplied.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

Read the picture in order. Retain the supported procedure facts; state that water-loss results were not supplied. The arrows show which recorded input or intermediate result each decision uses. Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

What this example does not establish

The examples are invented observations; no real classroom investigation or model run is claimed.

Read the labelled picture: What additional numbers would a half-water claim require? Point to the relevant step in the picture, then write the changed result and the rule or evidence that supports it.

Change one thing: E3 supplies losses of 1, 2 and 3 mL. Predict the result before checking the explanation. What remains the same?

7. Find and repair a mistake

ERROR DETECTIVE

Compare → Diagnose → Repair

Do not patch the answer. Find the earliest unsupported assumption, wrong step, or weak evidence.



COMPARE • DIAGNOSE

Weak idea: More features create a more complete first build. Strong move: inspect the claim, source, stage, schema, or test evidence; repair one cause; rerun.

Evidence record · Expected · Actual · Likely cause / next test

Evidence record: normal case

Expected: works

Actual: _____

Likely cause / next test: _____

Evidence record: boundary case

Expected: stated rule

Actual: _____

Likely cause / next test: _____

Evidence record: missing input

Expected: safe response

Actual: _____

Likely cause / next test: _____

Evidence record: fresh transfer

Expected: generalizes

Actual: _____

Likely cause / next test: _____

Repair one cause. Why is this repair better than changing several things?

Write or draw your reasoning and evidence.

A closer explanation

Complete the supported workbook tasks before attempting W4 independently. Record any hints or help used. After feedback, use the teacher's fresh retry rather than copying the earlier answer. For a project,

select a relevant new case, predict the outcome from the declared rule and preserve the result alongside any revision.

Exit: propose one revealing change to the worked scenario. State the expected outcome and the evidence that would make you revise your explanation.

8. Practise together

LOGIC STUDIO

Specify Before You Run

Prompts, schemas, retrieval, tool calls, and code are inspectable parts of one AI product.



SPECIFY THE SYSTEM

Define task, trusted context, constraints, schema, tools, checks, fallback, and stopping rule before trusting output.

SET task, audience, constraints, output_schema LOAD approved_context or retrieval_result CALL model or allowed_tool VALIDATE fields, claims, citations, and safety IF evidence missing: ask, retrieve, or abstain RETURN useful output + source trail + limitations

Trace one input. Which line changes the state or chooses a path?

Write or draw your reasoning and evidence.

A closer explanation

A correct happy path does not establish fallback, access control or side-effect safety.

A useful check asks what would make the reasoning fail. Distinguish a failure to execute the declared procedure from a weakness in the procedure itself. The first may require a correction; the second requires an explicitly revised method and new evidence. Preserve unsuccessful cases so an improved result cannot hide the original problem.

Guided: restate the worked result, then name the assumption or limitation that prevents a broader conclusion.

Practise with less help: Trace the worked case

Use workbook W2 for the connected example “Capstone grounded assistant”. First fill the input and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan · Do · Check

Plan: What is given? Which rule or procedure applies?

Do: Record each intermediate step.

Check: Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Check your reasoning

Use the worked case for Capstone grounded assistant. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

Answer on your own before discussion. If you are unsure, say which step you need help with.

Explain the picture

What additional numbers would a half-water claim require? Point to the relevant step in the picture, then write the changed result and the rule or evidence that supports it.

My explanation using a pictured step and the given facts

This is supported practice. Keep the separately named independent case for an attempt before feedback.

9. Run the investigation

EVIDENCE LAB

Predict → Test → Record → Explain

Keep expected and actual results separate. Surprises are useful data when recorded honestly.



PREDICT → RUN → VERIFY → EXPLAIN
Build the minimum flow, label components, link each to a requirement and test, then run a smoke test after every integration.

Build the minimum flow, label components, link each to a requirement and test, then run a smoke test after every integration.

Test / input · Prediction · Actual · Evidence / calculation · Change or keep?

Test / input: normal

Prediction: _____

Actual: _____

Evidence / calculation: _____

Change or keep?: _____

Test / input: boundary

Prediction: _____

Actual: _____

Evidence / calculation: _____

Change or keep?: _____

Test / input: missing

Prediction: _____

Actual: _____

Evidence / calculation: _____

Change or keep?: _____

Test / input: fresh transfer

Prediction: _____

Actual: _____

Evidence / calculation: _____

Change or keep?: _____

What did the hardest test teach you that the normal test did not?

Write or draw your reasoning and evidence.

Plan → try → check

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

10. Build your understanding

MASTERY

From Knowing the Word to Owning the Idea

Move upward only when you can produce evidence without copying the worked example.



CREATOR MASTERY

1 Define • 2 Trace • 3 Build • 4 Evaluate • 5 Verify • 6 Defend limits

- 1 • NAME: Define the concept without repeating the textbook.
- 2 • TRACE: Show the information, state, branch, calculation, or evidence path.
- 3 • APPLY: Solve a different example independently.
- 4 • CHALLENGE: Find a boundary case, counterexample, or unfair outcome.
- 5 • EXPLAIN LIMITS: Say when the result should not be trusted or used alone.

My strongest evidence of independent understanding:

Write or draw your reasoning and evidence.

11. Connect your project

CUMULATIVE PROJECT

Verified Study Buddy

Every week adds one inspectable piece. The system must show how it reached a result.



VERIFIED STUDY BUDDY

Assemble interface, retrieval, prompt, model/tool, validation, evidence display, and fallback.

THIS WEEK’S CHECKPOINT Assemble interface, retrieval, prompt, model/tool, validation, evidence display, and fallback.

SYSTEM MAP Learner need → prompt/schema → approved sources/retrieval → model/tool → structured answer → verification → human decision

What will you add, change, or verify in the project this week?

Write or draw your reasoning and evidence.

What should remain under human control?

Write or draw your reasoning and evidence.

RESPONSIBLE DESIGN Collect only necessary information. Show uncertainty and limits. Never confuse fluent output with verified truth.

Evidence for your project

For your Verified Study Buddy, save the input, procedure, observed result and limitation of this check. Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

12. Show what you know

INDEPENDENT MASTERY PROOF

Explain • Transfer • Test • Defend

Complete this page without copying the worked example. Your reasoning matters more than decoration.



INDEPENDENT CREATOR PROOF

Build the smallest complete Study Buddy workflow.

CHALLENGE Build the smallest complete Study Buddy workflow.

1. My fresh example, trace, calculation, or design:

Write or draw your reasoning and evidence.

2. My evidence that it works (include a boundary or missing-data case):

Write or draw your reasoning and evidence.

3. One limitation, fairness concern, or situation needing human review:

Write or draw your reasoning and evidence.

SELF-CHECK ☐ I explained the mechanism ☐ I used evidence ☐ I tested a fresh case ☐ I named a limitation ☐ I can answer 'why?'

13. Study a complete case

Week 34 Worked case

Use the supplied details to practise the weekly concept before attempting the independent question. This case extends the illustrated lesson.

The situation

A pipeline retrieves passages, drafts and displays without checking citations.

The question

What step is missing?

Worked explanation

Validate every factual claim against its citation, plus required fields and boundaries. Revise or stop unsupported answers; retain diagnostic stages.

Explain it in your own words

Which detail supports the answer, and why does it matter?

Write or draw your reasoning and evidence.

Trace the supplied case visually

Validate before displaying the draft

Evidence or stage · Value or result

Evidence or stage: Retrieve

Value or result: Relevant approved passages

Evidence or stage: Draft

Value or result: Compose the answer

Evidence or stage: Validate

Value or result: Check claims and citations

Evidence or stage: Display or stop

Value or result: Show supported content or explain the limit

A pipeline retrieves passages, drafts and displays without checking citations.

What step is missing?

Validate every factual claim against its citation, plus required fields and boundaries. Revise or stop unsupported answers; retain diagnostic stages.

14. Try a new case independently

Independent case: Changed independent case

An experiment note records four fictional seeds observed for six days but supplies no growth results. A

generated draft says all four sprouted and the method always works in any soil. Repair it.

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Week 34 Independent application

Close the worked explanation. Apply the idea to this different question without copying.

The interface looks complete, but its answer button returns the same saved sentence for every question. What capability is still missing, and how can you expose it?

My answer, with a trace, calculation or supporting evidence

Write or draw your reasoning and evidence.

Create and test

Change one relevant detail. Predict the result before testing. Include a boundary or missing-information case when it helps reveal a limit.

My changed input and expected result

Write or draw your reasoning and evidence.

Connect to your project

Assemble interface, retrieval, prompt, model/tool, validation, evidence display, and fallback.

The evidence I will keep

Write or draw your reasoning and evidence.

Your teacher has the independent answer and scoring criteria. Complete the matching workbook week to keep your practice evidence together.

Transfer practice from the original programme

A prototype cites note A but its answer adds a detail found nowhere in A. What should the final validation stage check and do?

Explain the mechanism, evidence and one limit.

Use feedback to improve

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed. A later check uses a changed case, not a copied answer.

15. Your lesson route

Start with this question: Build a claim checker for a draft summary of a fictional class investigation.

By the end, I can · How I will show it

By the end, I can: Explain the mechanism and terminology of capstone grounded assistant

How I will show it: Explain the deciding step, show a trace or test, and state one limit.

By the end, I can: Apply the stated procedure to a new case with a traceable result

How I will show it: Explain the deciding step, show a trace or test, and state one limit.

By the end, I can: Identify an assumption or limitation and design a revealing follow-up

How I will show it: Explain the deciding step, show a trace or test, and state one limit.

Remember → watch and explain → try with help → investigate → try independently → keep project evidence. Your teacher may spread this lesson across several classes.

Keep your first prediction. Compare it with what the evidence shows and explain any change.

16. Finish and return

Close your notes. Explain this goal in your own words: Explain the mechanism and terminology of capstone grounded assistant

My deciding step, evidence and one remaining question

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an input or

assumption and justify your prediction.

Save your workbook evidence. At the next review, try a different case before opening your old answer.

17. Supported practice, repair and portfolio

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Build a claim checker for a draft summary of a fictional class investigation.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? Retain the supported procedure facts; state that water-loss results were not supplied.

1. Extract claims
2. Match evidence
3. Repair draft

My selected conclusion, deciding fact and explanation

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

After your attempt, compare with the supported explanation: Repair draft Retain the supported procedure facts; state that water-loss results were not supplied. Read the picture in order. Retain the supported procedure facts; state that water-loss results were not supplied. The arrows show which recorded input or intermediate result each decision uses. Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Practice 2: explain the deciding evidence

Changed-case practice: E3 supplies losses of 1, 2 and 3 mL. Which conclusion is supported by the supplied facts?

1. We can decide the result without examining the changed input or the procedure.
2. Report those individual losses, not “half” unless starting volumes are also available.

3. This one example guarantees correct results for every future input.

My selected conclusion, deciding fact and explanation

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

After your attempt, compare with the supported explanation: Report those individual losses, not “half” unless starting volumes are also available. Absolute losses alone do not establish each cup’s fractional loss.

Repair a missing building block

Review Capstone Design (ai-3/33). Goal: Explain the mechanism and terminology of capstone requirements and corpus. Short practice: What should the project plan record before the new flower test? Point to the relevant step in the picture, then write the changed result and the rule or evidence that supports it.. Return to this week and explain what you changed.

Review Safety Boundaries (ai-3/31). Goal: Explain the mechanism and terminology of untrusted input boundaries. Short practice: If AB7 passes the pattern, may it access teacher-only material? Point to the relevant step in the picture, then write the changed result and the rule or evidence that supports it.. Return to this week and explain what you changed.

Review Privacy by Design (ai-3/28). Goal: Explain the mechanism and terminology of data retention and deletion. Short practice: Which IDs and locations must appear in the deletion evidence? Point to the relevant step in the picture, then write the changed result and the rule or evidence that supports it.. Return to this week and explain what you changed.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run evidence. Open the HTML copy to read or print it. Drafts stay in this browser when storage is available; clear drafts on a shared device.